

Empirical Research in Finance

Combined Course Submission · Group A

Humboldt-Universität zu Berlin
School of Business and Economics
Summer Semester 2026

PART I · GROUP SUBMISSION · EMPIRICAL ASSIGNMENT

Firm Valuation and Future Stock Returns

Evidence on the Value Effect and the Brexit Shock
among German Listed Firms

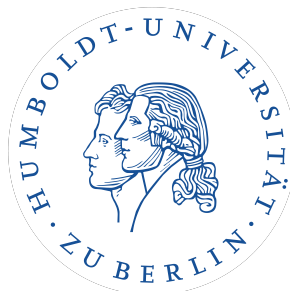
Julius Diesing

julius.diesing@student.hu-berlin.de
Business Administration
Bachelor of Science
Student ID 636509

Angeliki Tsekoura

angeliki.tsekoura@student.hu-berlin.de
Business Administration
Bachelor of Science
Student ID 633571

Course	Reference	Instructor	Submission
Empirical Research in Finance	701143	Dr. Beatriz Mariano	15 September 2026



Abstract

We examine valuation and one-year-ahead excess stock returns in a simulated panel of German listed firms. Higher book-to-market ratios are associated with higher returns, consistent with the Fama–French value pattern, and this relationship is stronger among small firms. It remains positive after including year and industry controls and clustering standard errors by firm. In the accompanying Brexit event study, UK sales exposure identifies substantially more negative abnormal returns, whereas UK listing status does not distinguish the groups’ reactions.

Data

The main analysis uses the simulated annual panel of German listed firms supplied through Moodle for this assignment. The file `value_effect_panel.dta` contains 3,060 firm-year observations for 255 firms from 2010 to 2021, linking characteristics in year t to excess returns in $t + 1$. This panel allows us to examine the value effect across firms and over time. The accompanying `cars_summary.dta` is used later for the Brexit event study and provides group-level abnormal returns and standard errors.

We clean the main panel by excluding financial firms (two-digit SIC 60–69), whose balance sheets and leverage differ from those of non-financial firms. We also require complete identifiers and regression inputs, meaning that the firm and year are known and none of the values needed to estimate the models is missing. The final sample contains 2,664 observations for 230 firms. It is an unbalanced panel because some firms do not have an observation in every year. Table 1 documents the filters.

Table 1. Sample selection

Sample restriction	Removed	Remaining
Original panel	–	3,060
Exclude financial firms (SIC 60–69)	300	2,760
Positive share price	10	2,750
Complete identifiers and regression inputs	86	2,664

Notes. Filters are applied sequentially. Each exclusion is counted once. Positive assets and shares ensure valid ratios. Neither check removes observations. Positive share prices also ensure valid log size. Complete inputs exclude missing values needed for identification or regression. Winsorisation removes no observations.

To limit the influence of extreme values, we winsorise excess returns, book-to-market, leverage, profitability and dividend yield at 1% in each tail across the final sample. This replaces the most extreme values with the respective boundary values without removing observations. Market capitalisation retains its original values to preserve the plausible size distribution. Its natural logarithm, our size measure, already compresses large differences and receives no further tail adjustment. Table 2 summarises the adjusted final sample used throughout the regressions.

Table 2. Final-sample descriptive statistics

Variable	Mean	SD	Min	P10	Median	P90	Max
Excess return	15.683	25.410	−44.288	−17.484	15.523	49.208	74.637
Book-to-market	1.409	0.704	0.100	0.452	1.400	2.357	2.898
Leverage	27.958	16.489	0.000	7.205	26.360	51.382	67.186
Profitability	7.549	7.540	−10.608	−2.217	7.523	17.533	25.816
Dividend yield	1.806	1.592	0.000	0.000	1.679	4.076	5.520
Size	3.981	1.839	−0.899	1.817	3.805	6.426	10.106
Market cap.	324.475	1,181.931	0.407	6.156	44.917	617.959	24,491.312

Notes. The sample contains 2,664 observations for 230 firms. Returns at $t + 1$, leverage, profitability and dividend yield are in percent. Book-to-market is a ratio. Market capitalisation is in EUR million. Size is its natural logarithm. The first five variables are winsorised at 1% in each tail. SD is the standard deviation. P10 and P90 are percentiles.

Annual excess returns average 15.683% and range from −44.288% to 74.637%, a wide but economically possible range for individual stocks. Book-to-market ranges from 0.100 to 2.898, covering valuations above and below book equity. Leverage of 0.000–67.186%, profitability of −10.608% to 25.816%, and dividend yields of 0.000–5.520% are plausible ranges that include loss-making and non-dividend-paying firms.

Market capitalisation ranges from EUR 0.407 million to EUR 24.491 billion. Its mean of EUR 324.475 million exceeds the EUR 44.917 million median, showing that a few very large firms pull the average well above the typical firm. This substantial right skew motivates the size controls used below. Log size ranges from −0.899 to 10.106. Negative log values simply indicate market capitalisation below EUR 1 million.

Methodology and Results

We begin with the baseline model in task (b), using one-year-ahead excess return as the dependent variable and book-to-market as the sole explanatory variable. Pooled OLS combines all firm-year observations in one regression. Every specification uses the same 2,664 observations, so changes in the results reflect changes in the model rather than the sample. The baseline book-to-market coefficient is positive and significantly different from zero (0.088, $t = 12.920$, $p < 0.001$). A one-unit higher ratio is associated with an 8.762-percentage-point higher excess return in the following year.

To assess whether this relationship also reflects other firm characteristics, model (c) adds size, leverage, profitability and dividend yield. For firm i in year t , the model is

$$r_{i,t+1}^e = \alpha + \beta_B \text{B2M}_{it} + \beta_S \text{Size}_{it} + \beta_L \text{Lev}_{it} + \beta_P \text{Prof}_{it} + \beta_D \text{DY}_{it} + \varepsilon_{it}. \quad (1)$$

Here, r^e is excess return, B2M is book-to-market, and *Size*, *Lev*, *Prof* and *DY* denote size, leverage, profitability and dividend yield. The intercept is α and ε is the error term. Returns and financial ratios enter in decimal form. The book-to-market coefficient falls to 0.059 but remains significant ($p < 0.001$), suggesting that the baseline relationship partly overlaps with the added characteristics. Size is negatively associated with returns ($p < 0.001$), while profitability is positively associated ($p = 0.036$). Leverage and dividend yield are not individually significant at 5% ($p = 0.532$ and $p = 0.115$). We retain them to account for financing and payout differences when estimating the other coefficients. A non-significant individual test does not establish that a variable is irrelevant as a control. Table 3 consolidates all regression results.

Table 3. One-year-ahead excess returns

	(b)	(c)	(d)/(e)	(f)	(g)
Book-to-market	0.088*** (0.007)	0.059*** (0.008)	0.043*** (0.010)	0.044*** (0.008)	0.044*** (0.010)
Size		-0.023*** (0.003)			
Small firm			0.033 (0.022)	0.034* (0.019)	0.034* (0.020)
Book-to-market $\times$ Small			0.046*** (0.014)	0.045*** (0.012)	0.045*** (0.013)
Leverage		-0.018 (0.029)	-0.025 (0.029)	-0.028 (0.025)	-0.028 (0.024)
Profitability		0.149** (0.071)	0.161** (0.071)	0.131** (0.062)	0.131** (0.056)
Dividend yield		0.529 (0.336)	0.330 (0.336)	0.500* (0.288)	0.500* (0.290)
Constant	0.033*** (0.011)	0.150*** (0.022)	0.031* (0.017)	-0.139*** (0.024)	-0.139*** (0.025)
Observations	2,664	2,664	2,664	2,664	2,664
R^2	0.059	0.085	0.099	0.349	0.349
Adjusted R^2	0.059	0.083	0.097	0.343	0.343
Year dummies	No	No	No	Yes	Yes
Industry dummies	No	No	No	Yes	Yes
Standard errors	OLS	OLS	OLS	OLS	Clustered

Notes. The dependent variable is excess return at $t+1$ in decimal form. All models use the same 230 firms. Standard errors are in parentheses. Column (g) uses HC1 firm-clustered standard errors and 229 degrees of freedom for t -tests. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Task (d) asks whether the value effect differs between small and large firms. We replace continuous size with $Small_{it}$, which equals one when market capitalisation is below the full-sample median of EUR 44.917 million and zero otherwise. This common threshold is calculated across all firm-year observations, so a firm can change classification over time. Table 4 reports the annual counts.

Table 4. Small-firm observations by year

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
N	116	118	117	120	118	117	109	110	105	105	99	98

Notes. Small firms have market capitalisation below the full-sample median of EUR 44.917 million.

The dummy captures a difference in return levels, while its interaction with book-to-market allows a different slope for small firms. We estimate

$$r_{i,t+1}^e = \alpha + \beta_B B2M_{it} + \delta Small_{it} + \gamma(B2M_{it} \times Small_{it}) + \beta_L Lev_{it} + \beta_P Prof_{it} + \beta_D DY_{it} + \varepsilon_{it}. \quad (2)$$

The book-to-market slope is β_B for large firms and $\beta_B + \gamma$ for small firms. The null hypothesis $H_0: \gamma = 0$ therefore states that the slopes are equal, while $H_1: \gamma \neq 0$ states that they differ. We reject the null ($t = 3.252$, $p = 0.001$). A one-unit higher book-to-market ratio is associated with returns 4.259 percentage points higher among large firms and 8.874 points higher among small firms. The difference of 4.615 points indicates an economically stronger value effect among small firms, although the regression cannot distinguish risk compensation from mispricing.

Having established a difference between the slopes, task (e) tests whether book-to-market is unrelated to returns in both groups. This requires the joint null $H_0: \beta_B = \gamma = 0$, against $H_1: \beta_B \neq 0$ or $\gamma \neq 0$. Comparing equation (2) with a restricted model excluding both book-to-market terms gives $F(2, 2657) = 47.512$ ($p < 0.001$). We reject the joint null. This agrees with the positive relationship in model (c), which imposed one common book-to-market slope. The interaction model shows that this common slope conceals a stronger relationship among small firms. The restricted model and test are included in the Appendix.

Differences across calendar years and industries could also influence the estimated relationship. Model (f) therefore adds year effects λ_t and two-digit SIC industry effects $\eta_{s(i,t)}$,

$$r_{i,t+1}^e = \alpha + \beta_B B2M_{it} + \delta Small_{it} + \gamma(B2M_{it} \times Small_{it}) + \beta_L Lev_{it} + \beta_P Prof_{it} + \beta_D DY_{it} + \lambda_t + \eta_{s(i,t)} + \varepsilon_{it}. \quad (3)$$

Year effects capture common annual shocks, while industry effects capture average return differences across industries. The model includes 11 year dummies and nine industry dummies, with 2010 and SIC 20 as the respective reference categories. Both book-to-market coefficients remain close to column (d)/(e) and significant at 1%. Adjusted R^2 rises from 0.097 to 0.343, and the added dummies are jointly significant ($F(20, 2637) = 50.607, p < 0.001$). This supports the richer specification because it explains more return variation even after accounting for the additional regressors. It does not, however, establish a causal effect of valuation.

Finally, column (g) uses HC1 standard errors clustered by firm. This allows the error variance to differ across observations and errors for the same firm to be correlated over time. Coefficients and model fit are unchanged because clustering changes only the estimated uncertainty. Both book-to-market terms remain significant at 1%. Profitability is positive and significant ($\hat{\beta}_P = 0.131, t = 2.335, p = 0.020$). Using the unrounded estimate, a ten-percentage-point increase in profitability corresponds to $\hat{\beta}_P \times 0.100 \times 100 \approx 1.311$ percentage points higher subsequent excess returns, holding the other regressors fixed.

Event Study

We now turn from annual return differences to the market reaction around a specific event. The event date is Friday, 24 June 2016, when the Brexit referendum result became known before trading. The $[-1, +1]$ window covers 23, 24 and 27 June, and $[-2, +2]$ covers 22, 23, 24, 27 and 28 June. Both exclude the intervening weekend.

Using 2015 classifications, UK sales exposure means that at least 10% of sales originate in the UK. The alternative treatment is a UK listing. The final panel contains 54 sales-exposed and 176 control firms, compared with 40 UK-listed and 190 control firms. Table 5 shows that the definitions disagree for 74 firms (32.174%).

Table 5. UK exposure classifications in the final sample

UK sales share	Not UK-listed	UK-listed	Total
Below 10%	146	30	176
At least 10%	44	10	54
Total	190	40	230
Classified differently			74

We use the supplied group-level average abnormal returns and standard errors. Cumulative abnormal returns sum daily abnormal returns. Their standard errors equal the daily standard error times $\sqrt{L}$ for L trading days, assuming no correlation across days. For independent treated and control groups, $SE_{\text{diff}} = \sqrt{SE_T^2 + SE_C^2}$. Test statistics divide each return or difference by its standard error. Table 6 reports two-sided normal-approximation tests under both definitions.

Table 6. Abnormal returns around the Brexit referendum result

Definition	Window	Treated	Control	Difference
UK sales	Event day	-4.678*** (-28.682)	-0.032 (-0.371)	-4.646*** (-25.108)
	[-1, +1]	-4.878*** (-17.267)	0.097 (0.638)	-4.975*** (-15.522)
	[-2, +2]	-4.927*** (-13.509)	-0.065 (-0.331)	-4.862*** (-11.752)
UK listing	Event day	-1.040*** (-5.538)	-1.121*** (-13.278)	0.081 (0.395)
	[-1, +1]	-0.949*** (-2.918)	-1.076*** (-7.359)	0.127 (0.357)
	[-2, +2]	-1.051** (-2.502)	-1.219*** (-6.453)	0.168 (0.364)

Notes. Group returns are in percent. Differences are treated minus control in percentage points. Test statistics are in parentheses. Event-day values are average abnormal returns. Longer windows report cumulative average abnormal returns. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Sales-exposed firms lose 4.678% on the event day, compared with 0.032% for controls. The treated-minus-control difference is -4.646 percentage points and remains similar over three and five days (-4.975 and -4.862 points). All differences are significant at 1%, whereas control-group returns are insignificant. Under the listing definition, both groups lose roughly 1%, but none of their differences is significant.

UK sales capture exposure to UK demand and revenues, whereas a UK listing identifies a trading venue. The conclusion about differential losses therefore depends on the exposure measure. The supplied return file has no firm identifiers, so its aggregates cannot be recalculated for the filtered panel.

Conclusion

The simulated panel displays a positive value effect that is stronger among small firms and remains significant with year and industry controls and firm-clustered standard errors. Profitability is also positively associated with subsequent returns. These patterns are consistent with familiar asset-pricing intuition, but simulated data cannot independently validate a real-world return premium. The Brexit results favour UK sales over listing status as a measure of operating exposure, showing why the economic meaning of a treatment definition matters for the conclusion.

Appendix

The complete public repository contains the reports, data, R code and pinned package versions: <https://github.com/JuliusD9500/empirical-research-in-finance-2026>.

Analyses were conducted in R. Tables were formatted separately in LaTeX.

Run the scripts from the repository root in the order below, with both supplied data files in `data/raw/`. The setup restores the versions recorded in `renv.lock`. The analysis scripts display results in R. The report tables are maintained separately in `paper/group/tables/` and are not overwritten when the analysis is run.

A.1 Environment setup

```
# =====  
# Empirical Research in Finance - initial setup  
#  
# Purpose: Restore the project packages at their saved versions.  
# Input: renv.lock (run this script from the repository root)  
# Output: Isolated project library in renv/library/  
# =====  
  
renv_installed <- requireNamespace("renv", quietly = TRUE)  
  
if (!renv_installed) { # ! is logical not  
  install.packages("renv", repos = "https://cloud.r-project.org")  
}  
  
renv::load() # use this project's isolated package library  
renv::restore(prompt = FALSE) # install the versions in renv.lock
```

A.2 Data preparation

```
# =====  
# Empirical Research in Finance - Group Assignment  
#  
# Purpose: Filter the sample, create variables and winsorize.  
# Input: data/raw/value_effect_panel.dta  
# Output: data/clean/value_effect_panel_clean.rda (final sample)  
# =====  
  
library(tidyverse) # data wrangling  
library(labelled) # variable labels  
  
# Load raw data  
data <- haven::read_dta("data/raw/value_effect_panel.dta") |>  
  haven::zap_label()  
  
# Filter data (balance sheet identity is violated; german firms only, no  
  country filter needed)  
data <- data |>  
  filter(  
    !(sic2 >= 60 & sic2 <= 69), # exclude financial firms  
    total_assets > 0,           # for 'lev' and 'prof'  
    num_shares > 0,             # for 'market_cap'  
    share_price > 0,           # for 'market_cap' and 'size'  
    complete.cases(  
      firm_id, year, sic2, excess_return,  
      total_assets, num_shares, share_price,  
      book_equity, lt_debt, debt_current, ebitda, dividends_total  
    )  
  )  
  
# Create variables  
data <- data |>  
  mutate( # keep original and add new variables  
    market_cap = share_price * num_shares,  
    b2m = book_equity / market_cap,  
    lev = (lt_debt + debt_current) / total_assets,  
    prof = ebitda / total_assets,  
    div_yield = dividends_total / market_cap,  
    size = log(market_cap)  
  )  
  
# Winsorize variables (1% in each tail, no interpolation as in the lecture)  
winsorize <- function(x) {  
  sorted_x <- sort(x)  
  n <- length(sorted_x)  
  k <- floor(0.01 * n)  
  
  lower <- sorted_x[k + 1]
```

```

upper <- sorted_x[n - k]

x[x < lower] <- lower
x[x > upper] <- upper

return(x)
}

data <- data |> # keep original variables and add winsorized ones
mutate(
  excess_return_w = winsorize(excess_return),
  b2m_w = winsorize(b2m),
  lev_w = winsorize(lev),
  prof_w = winsorize(prof),
  div_yield_w = winsorize(div_yield),
  # not size bc. logarithm weakens large variations (span is plausibly)
)

# Label variables
data <- data |>
labelled::set_variable_labels(
  firm_id      = "Firm identifier",
  year         = "Calendar year",
  sic2         = "Industry (2-digit SIC)",

  total_assets = "Total assets (EUR mn)",
  book_equity  = "Book equity (EUR mn)",
  lt_debt      = "Long-term debt (EUR mn)",
  debt_current = "Debt in current liabilities (EUR mn)",
  total_liabilities = "Total liabilities (EUR mn)",
  ebitda       = "EBITDA (EUR mn)",
  num_shares   = "Shares outstanding (mn)",
  share_price  = "Year-end share price (EUR)",
  dividends_total = "Total dividends (EUR mn)",
  total_sales  = "Net sales (EUR mn)",
  rf           = "Risk-free rate (decimal)",
  excess_return = "Excess return at t+1 (decimal)",

  uk_exposure  = "UK sales share >= 10% in 2015",
  uk_listed    = "UK listing in 2015",

  # Constructed variables
  market_cap   = "Market capitalisation (EUR mn)",
  b2m           = "Book-to-market (ratio)",
  lev           = "Leverage (ratio)",
  prof          = "Profitability (ratio)",
  div_yield     = "Dividend yield (ratio)",
  size          = "Firm size (ln market capitalisation in EUR mn)",

  # Winsorized variables

```

```
    excess_return_w = "Excess return at t+1 (winsorized, decimal)",
    b2m_w           = "Book-to-market (winsorized, ratio)",
    lev_w           = "Leverage (winsorized, ratio)",
    prof_w          = "Profitability (winsorized, ratio)",
    div_yield_w     = "Dividend yield (winsorized, ratio)"
)

# Save cleaned data
save(data, file = "data/clean/value_effect_panel_clean.rda")
```

A.3 Descriptive statistics – task (a)

```
# =====  
# Empirical Research in Finance - (a)  
#  
# Purpose: Describe the final sample, including units and sample size.  
# Input: data/clean/value_effect_panel_clean.rda from  
#        code/01_data_preparation.R  
# Output: Descriptive statistics table (displayed in R)  
# =====  
  
# %% Load required packages  
library(tidyverse) # data wrangling  
library(modelsummary) # descriptive statistics table  
  
# %% Load cleaned data  
load("data/clean/value_effect_panel_clean.rda")  
  
# %% Choose features for task  
task_a_df <- data |>  
  haven::zap_label() |>  
  transmute(  
    `Excess return`1 2` = excess_return_w * 100,  
    `Book-to-market`2` = b2m_w,  
    `Leverage`2` = lev_w * 100,  
    `Profitability`2` = prof_w * 100,  
    `Dividend yield`2` = div_yield_w * 100,  
    `Size` = size,  
    `Market capitalisation` = market_cap  
  )  
  
# Define function for percentiles  
percentile <- function(x, p) {  
  return(quantile(x, probs = p))  
}  
  
# Calculate 10th and 90th percentiles  
P10 <- function(x) percentile(x, 0.1) # reminder: Python equivalent to lambda  
  x: ...  
P90 <- function(x) percentile(x, 0.9)  
  
# Units-Column for descriptive table  
table_units <- data.frame(  
  Unit = c("%", "ratio", "%", "%", "%", "ln(EUR mn)", "EUR mn")  
)  
attr(table_units, "position") <- 2  
  
# Final sample size  
n_observations <- nrow(data)  
n_firms <- n_distinct(data$firm_id)
```

```

# Descriptive statistics table
table_a <- datasummary(
  All(task_a_df) ~ Mean + SD + Min + P10 + Median + P90 + Max,
  data = task_a_df,
  add_columns = table_units,
  fmt = 3,
  align = "llrrrrrrr",
  title = "Final-sample descriptive statistics",
  notes = paste("Observations:", n_observations, "|| Firms:", n_firms, "|| 1
    Returns refer to t+1 || 2 Winsorized variables at 1% in each tail || All
    statistics use the single final sample"),
  output = "tinytable"
)

# Display tiny table
table_a

```

A.4 Regressions and tests – tasks (b)–(g)

```
# =====
# Empirical Research in Finance - (b)-(g)
#
# Purpose: Estimate the return models and test the value effect.
# Input: data/clean/value_effect_panel_clean.rda from
#        code/01_data_preparation.R
# Output: Tests and tables (displayed in R)
# =====

# %% Load required packages
library(dplyr) # data wrangling
library(sandwich) # covariance matrix
library(lmtest) # coefficient tests
library(modelsummary) # regression tables

# %% Load cleaned data
load("data/clean/value_effect_panel_clean.rda") # name: 'data'

# %% (b) Pooled OLS regression
model_b <- lm(
  excess_return_w ~ b2m_w,
  data = data
)
print(summary(model_b))

# Test coefficient: H0:  $\beta_B = 0$ ; H1:  $\beta_B \neq 0$ . Reject H0 because  $p < 0.05$ 
test_b <- coef(summary(model_b))["b2m_w", ]
print(test_b)

# %% (c) Add controls
model_c <- lm(
  excess_return_w ~ b2m_w + size + lev_w + prof_w + div_yield_w,
  data = data
)
print(summary(model_c))

# Test controls: H0:  $\beta_j = 0$ ; H1:  $\beta_j \neq 0$ 
test_c <- coef(summary(model_c))[c("size", "lev_w", "prof_w", "div_yield_w"), ]
print(test_c)

# %% (d) Small firms and the value effect
data <- data |>
  dplyr::mutate(small_firm = as.numeric(market_cap < median(market_cap)))

# Estimated model with interaction term
model_d <- lm(
  excess_return_w ~ b2m_w * small_firm + lev_w + prof_w + div_yield_w,
```

```

    data = data
  )
print(summary(model_d))

# Test interaction: H0: gamma = 0; H1: gamma != 0
test_d <- coef(summary(model_d))["b2m_w:small_firm", ]
print(test_d)

# Annual counts table
table_d <- datasummary(
  Factor(year, name = "Year") ~ N, #left: rows, right: column -> number of
    observations
  data = data |> dplyr::filter(small_firm == 1),
  fmt = 0,
  align = "lr",
  title = "Small-firm observations by year",
  notes = "Small if market capitalisation is below the full-sample median",
  output = "tinytable"
)

# Display table
table_d

# Book-to-market slopes by firm size (percentage points)
slopes_d_pp <- c(
  large = coef(model_d)[["b2m_w"]],
  small = coef(model_d)[["b2m_w"]] + coef(model_d)[["b2m_w:small_firm"]],
  diff = coef(model_d)[["b2m_w:small_firm"]]
) * 100 # economic interpretation
print(slopes_d_pp)

# %% (e) Estimate the model without interaction term
model_e <- lm(
  excess_return_w ~ small_firm + lev_w + prof_w + div_yield_w,
  data = data
)

# Joint F-test of no value effect (value companys: high b2m + high dividend /
  growth companies: low b2m + low dividend)
# H0: beta_B = gamma = 0; H1: at least one is nonzero.
test_e <- anova(model_e, model_d)
print(test_e)

# %% (f) Add year and industry dummies
model_f <- lm(
  excess_return_w ~ b2m_w * small_firm + lev_w + prof_w + div_yield_w +
    factor(year) + factor(sic2),
  data = data
)

```

```

print(summary(model_f))

# Test dummies: H0: all added coefficients = 0; H1: at least one is nonzero.
test_f <- anova(model_d, model_f)
print(test_f)

# %% (g) Firm-clustered standard errors
n_firms <- n_distinct(data$firm_id)
vcov_g <- vcovCL(model_f, cluster = ~ firm_id, type = "HC1")
model_g <- coeftest(model_f, vcov. = vcov_g, df = n_firms - 1)

# Test profitability: H0:  $\beta_P = 0$ ; H1:  $\beta_P \neq 0$ .
test_g <- model_g["prof_w", ]
print(test_g)

# Return change for a 10-percentage-point increase in profitability
  (percentage points)
profitability_effect_pp <- coef(model_f)["prof_w"] * 0.10 * 100
print(profitability_effect_pp)

# %% Regression table
# Combine clustered inference from (g) with model fit from (f)
model_g_table <- modelsummary(model_g, output = "modelsummary_list")
model_g_table$glance <- get_gof(model_f)

# Additional rows below the regression results
table_details <- data.frame(
  term = c("Year dummies", "Industry dummies", "Standard errors", "Firms"),
  "(b)" = c("No", "No", "OLS", n_firms),
  "(c)" = c("No", "No", "OLS", n_firms),
  "(d)/(e)" = c("No", "No", "OLS", n_firms),
  "(f)" = c("Yes", "Yes", "OLS", n_firms),
  "(g)" = c("Yes", "Yes", "Clustered", n_firms),
  check.names = FALSE
)

# Consolidated regression table; (e) tests the model from (d)
table_b_g <- modelsummary(
  list("(b)" = model_b, "(c)" = model_c, "(d)/(e)" = model_d,
       "(f)" = model_f, "(g)" = model_g_table),
  coef_map = c( # select, label and order coefficients; omit year/industry
                dummies
                "b2m_w" = "Book-to-market",
                "size" = "Size",
                "small_firm" = "Small firm",
                "b2m_w:small_firm" = "Book-to-market x Small firm",
                "lev_w" = "Leverage",
                "prof_w" = "Profitability",
                "div_yield_w" = "Dividend yield",
                "(Intercept)" = "Constant"
  )

```

```

),
gof_map = c("nobs", "r.squared", "adj.r.squared"),
add_rows = table_details,
statistic = "{std.error}",
stars = c("*" = 0.10, "**" = 0.05, "***" = 0.01),
fmt = 3,
align = "lrrrrr",
width = c(0.32, 0.136, 0.136, 0.136, 0.136, 0.136),
title = "One-year-ahead excess returns",
notes = paste(
  "Dependent variable: excess return at t+1 (decimal). All models use the
  same final sample.",
  "Standard errors in parentheses. Column (g): HC1 firm-clustered SEs;
  t-tests use firms minus 1 degrees of freedom."
),
output = "tinytable"
)

# Display regression table
table_b_g

```

A.5 Event study – task (h)

```
# =====  
# Empirical Research in Finance - Group Assignment  
#  
# Purpose: A comparison of the stock price reaction to the Brexit referendum  
#          between UK-exposed and non-exposed German firms.  
# Input:  
# Output:  
# =====  
library(tidyverse)  
library(haven)  
  
#Load cleaned firm panel  
load("data/clean/value_effect_panel_clean.rda")  
  
#Load event-study summary file (no cleaning needed)  
cars <- haven::read_dta("data/raw/cars_summary.dta") |> haven::zap_label()  
  
# 1. Event design  
  
event_date <- as.Date("2016-06-24") # Brexit announcement  
event_windows <- list(  
  "[-1,+1]" = c(start = as.Date("2016-06-23"), end = as.Date("2016-06-27")),  
  "[-2,+2]" = c(start = as.Date("2016-06-22"), end = as.Date("2016-06-28"))  
)  
  
# The event date is Friday, 24 June 2016. The [-1,+1] window covers  
# 23, 24 and 27 June; [-2,+2] also includes 22 and 28 June.  
# The weekend of 25-26 June is excluded from both trading-day windows.  
  
# 2. Sample composition  
  
# sanity check: uk_exposure / uk_listed should be fixed per firm  
data |>  
  group_by(firm_id) |>  
  summarise(n_exp = n_distinct(uk_exposure), n_listed = n_distinct(uk_listed))  
  |>  
  filter(n_exp > 1 | n_listed > 1) # should return 0 rows  
  
firm_level <- data |>  
  distinct(firm_id, uk_exposure, uk_listed)  
  
# counts under each definition  
firm_level |> count(uk_exposure)  
firm_level |> count(uk_listed)  
  
# cross-tabulation  
cross_tab <- firm_level |>  
  count(uk_exposure, uk_listed) |>
```

```

pivot_wider(names_from = uk_listed, values_from = n, values_fill = 0)
cross_tab

# firms classified differently across the two definitions
n_different <- firm_level |>
  filter(uk_exposure != uk_listed) |>
  nrow()
n_different

# Results explained

# Final sample: 230 firms (2664 firm-year observations)
#
# Treated/control counts under each definition:
# UK sales exposure: ( dummy variable -> if the firm gets 10% of its sales
  from the UK. 1 = yes (exposed), 0 = no.): 54 treated (1), 176 control (0)
# UK listing status: ( dummy variable -> the firm's shares are also traded
  on a UK stock exchange. 1 = yes (listed), 0 = no.) 40 treated (1), 190
  control (0)
# 4 possible pairings: (0,0), (0,1), (1,0), (1,1)

# Cross-tabulation (UK sales exposure x UK listing status):
#
#
#           UK not listed (0)   UK listed (1)   Total
# UK sales exp < 10% (0)      146 (0,0)        30 (0,1)    176
# UK sales exp >= 10% (1)     44 (1,0)         10 (1,1)     54
# Total                       190              40          230

# The top-left cell with the number 146 means that 146 firms have less than
  10% UK sales and are not UK-listed -> firms with no UK connection under
  either measure.
# The bottom-right cell with the number 10 means that 10 firms have both 10%
  UK sales and are UK-listed -> firms connected to the UK under both
  measures.
# 74 (30+44) (32% of the sample)
# 30 firms are UK-listed but do not meet the 10% sales-exposure threshold
# 44 firms have >=10% UK sales exposure but are not UK-listed.
# This indicates the two definitions disagree.
# aspects of a firm's UK relationship (revenue exposure vs.listing venue),
  not the same underlying concept.

# (3) Statistical tests -- UK sales exposure

# CAR = cumulative abnormal return, SE = standard error,
# t-stat = CAR / SE, sig = significance stars based on the t-stat

windows <- list(
  "Event day" = 0,
  "[-1,+1]"   = -1:1,
  "[-2,+2]"   = -2:2

```

```

)

# sum daily AARs into CAR, scale SE by sqrt(window length), and get
# treated/control/difference. for the difference row we add variances
# (not SEs directly) since treated and control are independent groups
car_table_exposure <- map_dfr(names(windows), function(win_name) {
  win_days <- windows[[win_name]]

  cars |>
    filter(def == "uk_exposure", evttime %in% win_days) |>
    group_by(group) |>
    summarise(
      CAR = sum(aar_m),
      SE = unique(aar_se) * sqrt(length(win_days)),
      .groups = "drop"
    ) |>
    pivot_wider(names_from = group, values_from = c(CAR, SE)) |>
    mutate(
      CAR_difference = CAR_treated - CAR_control,
      SE_difference = sqrt(SE_treated^2 + SE_control^2),
      window = win_name
    )
}) |>
pivot_longer(
  cols = c(CAR_treated, CAR_control, CAR_difference, SE_treated, SE_control,
    SE_difference),
  names_to = c(".value", "group"),
  names_pattern = "(CAR|SE)_(.*)"
) |>
mutate(
  t_stat = CAR / SE,
  sig = case_when(
    2 * (1 - pnorm(abs(t_stat))) < 0.01 ~ "***",
    2 * (1 - pnorm(abs(t_stat))) < 0.05 ~ "**",
    2 * (1 - pnorm(abs(t_stat))) < 0.10 ~ "*",
    TRUE ~ ""
  )
) |>
select(window, group, CAR, SE, t_stat, sig)

car_table_exposure

# results:
# treated firms drop around 4.7-4.9% in every window and it's super
# significant (t around -13 to -29). control firms barely move at all
# and aren't significant. the difference between the two groups is
# also huge and significant (t around -12 to -25), so UK sales
# exposure really does seem to explain a big chunk of the reaction

# =====

```

```

# (4) Robustness check -- UK listing status
# =====
# same test as (3), just swapping in the UK listing dummy instead
car_table_listed <- map_dfr(names(windows), function(win_name) {
  win_days <- windows[[win_name]]

  cars |>
    filter(def == "uk_listed", evttime %in% win_days) |>
    group_by(group) |>
    summarise(
      CAR = sum(aar_m),
      SE = unique(aar_se) * sqrt(length(win_days)),
      .groups = "drop"
    ) |>
    pivot_wider(names_from = group, values_from = c(CAR, SE)) |>
    mutate(
      CAR_difference = CAR_treated - CAR_control,
      SE_difference = sqrt(SE_treated^2 + SE_control^2),
      window = win_name
    )
}) |>
  pivot_longer(
    cols = c(CAR_treated, CAR_control, CAR_difference, SE_treated, SE_control,
              SE_difference),
    names_to = c(".value", "group"),
    names_pattern = "(CAR|SE)_(.*)"
  ) |>
  mutate(
    t_stat = CAR / SE,
    sig = case_when(
      2 * (1 - pnorm(abs(t_stat))) < 0.01 ~ "***",
      2 * (1 - pnorm(abs(t_stat))) < 0.05 ~ "**",
      2 * (1 - pnorm(abs(t_stat))) < 0.10 ~ "*",
      TRUE ~ ""
    )
  ) |>
  select(window, group, CAR, SE, t_stat, sig)

car_table_listed

# results:
# this time both treated AND control firms drop by about the same
# amount (around 1-1.2%), and both are significant on their own. but
# the difference between them is tiny and never significant (t only
# around 0.36-0.40). so being UK-listed doesn't really separate firms
# that got hit harder from ones that didn't -- unlike sales exposure,
# this definition doesn't seem to capture real Brexit exposure

```

Empirical Research in Finance

Combined Course Submission · Group A

Humboldt-Universität zu Berlin
School of Business and Economics
Summer Semester 2026

PART II · INDIVIDUAL SUBMISSION · RESEARCH PROPOSAL

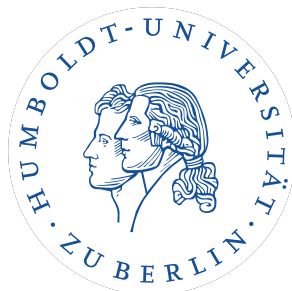
Cash Holdings and Corporate Investment

Capital Expenditure after Hurricane Helene
Evidence from US Manufacturing Firms

Julius Diesing

julius.diesing@student.hu-berlin.de
Business Administration
Bachelor of Science
Student ID 636509

Course	Reference	Instructor	Submission
Empirical Research in Finance	701143	Dr. Beatriz Mariano	15 September 2026



Background

A natural disaster can increase firms' need to replace machinery and facilities while disrupting the revenues available to finance investment. Cash accumulated before the event may help bridge this funding gap. Hurricane Helene's impact on the southeastern United States in September 2024 provides a setting to examine this financing problem.

Duchin, Ozbas and Sensoy (2010) find that corporate investment declined more strongly among firms with lower cash reserves during the financial crisis beginning in 2007. Their findings suggest that internal funds can help maintain investment when external finance becomes harder to obtain. This proposal examines whether a similar relationship arises after a physical shock that also changes firms' investment needs.

The research question is: among manufacturing firms domiciled and listed in the United States, is the change in capital expenditure associated with exposure to Helene more positive for firms with higher pre-storm cash holdings? I would compare exposed and unexposed firms before the storm and during the first four full quarters afterwards. This separates the response associated with exposure from a general tendency for firms with more cash to invest more.

Expected Outcome

My main hypothesis (H_1) is that the change in capital expenditure associated with exposure to Helene is more positive for firms with higher pre-storm cash holdings. Cash can finance replacement assets and maintain planned investment when operating receipts fall. It can also cover the period before external finance or insurance payments become available.

Firms with more cash may increase capital expenditure more or reduce it less, so H_1 does not require an overall rise in investment. Insurance and access to credit may reduce the importance of cash. Conversely, shortages of labour or equipment may delay investment even when funds are available.

Data

I would use a quarterly panel of US-domiciled, US-listed manufacturing firms (SIC 2000 to 3999) with December fiscal year ends and verifiable US production locations. Manufacturing provides a direct link to physical assets and their replacement. The sample would cover eight pre-storm quarters (2022Q3 to 2024Q2) and four post-storm quarters (2024Q4 to 2025Q3). Two prior annual cycles would help assess investment trends, while the following year would capture initial responses. I would exclude 2024Q3 so that each observation covers a quarter entirely before or after Helene.

Financial data would come from Compustat North America through WRDS, subject to obtaining access. Company annual reports available through the SEC would provide production locations reported before Helene. The Federal Emergency Management Agency (FEMA) supplies disaster declarations identifying counties approved for assistance after the storm.

Compustat's CAPXY records capital expenditure accumulated within the fiscal year. I would calculate quarterly capex by subtracting the previous quarter's accumulated amount within that year. The first quarter needs no subtraction. Investment (I_{it}) would equal 100 times quarterly capex divided by total assets at the end of 2023. This fixed denominator prevents the ratio from rising simply because reported assets fall after the storm. Cash holdings (C_i) would equal cash and short-term investments (CHEQ) divided by total assets (ATQ), both measured at the end of 2023, before Helene could affect the balance sheet.

The exposure dummy (E_i) would equal one if at least one principal US production facility is in a county designated for FEMA Individual Assistance following Helene, and zero otherwise. This measures geographic exposure rather than confirmed firm damage. Firms without such facilities would form the control group. I would check whether both groups include comparable industries and have similar pre-storm investment patterns. Production locations are used because headquarters may be elsewhere.

I would require positive assets at the end of 2023 and exclude firms with incomplete location records or overlapping facility exposure to other major storms in 2024. I would check mergers and divestitures for changes in reporting scope. Firms leaving the market would remain for available periods, and missing observations would not be replaced by zero.

Methodology

I would estimate a difference-in-differences model by OLS to compare investment changes in exposed and unexposed firms. Here, i identifies firms and t identifies quarters. The dependent variable is investment I_{it} , defined above. The explanatory variables combine exposure E_i , initial cash holdings C_i and a dummy Post_t equal to one from 2024Q4 onwards and zero before Helene:

$$I_{it} = \alpha_i + \lambda_t + \beta_1(E_i \times \text{Post}_t) \\ + \beta_2(\text{Post}_t \times C_i) + \beta_3(E_i \times \text{Post}_t \times C_i) + \varepsilon_{it}.$$

Firm fixed effects (α_i) account for persistent differences between firms. Quarter fixed effects (λ_t) capture shocks shared across firms in each quarter, and ε_{it} is the error term. The simpler terms E_i , C_i and $E_i \times C_i$ are absorbed by firm effects because they do not change over time. Quarter effects absorb Post_t . Thus, all lower-order terms underlying the three-way interaction are included or absorbed.

The term $\text{Post}_t \times C_i$ allows cash to be associated with investment changes even among unexposed firms. Consequently, the exposure-related change at cash level C_i is $\beta_1 + \beta_3 C_i$. The coefficient β_3 answers the research question: a ten-percentage-point difference in cash/assets implies a difference of $0.100\beta_3$ percentage points in the quarterly investment response.

I would test $H_0 : \beta_3 \leq 0$ against $H_1 : \beta_3 > 0$ using a one-sided t-test with standard errors clustered by firm, and report a 95% confidence interval. The comparison assumes that, without Helene, exposed and unexposed firms would have followed comparable investment trends at similar cash levels. Pre-storm plots for both groups, separated by initial cash levels, would help assess this assumption without proving it.

Extensions and Limitations

One limitation is that firms choose their cash reserves. Firms with more cash may also be better prepared or insured, which can independently affect investment after Helene. Firm fixed effects do not fully separate these influences. A positive β_3 would support an association between cash and investment, but would not establish a causal cash effect.

As an extension, I would replace cash holdings with net short-term liquidity, measured as $(\text{CHEQ} - \text{DLCQ})/\text{ATQ}$ at the end of 2023. DLCQ records debt in current liabilities. Subtracting these obligations accounts for funds potentially needed for repayment. I would retain negative values and use the same sample, outcome and regression. This would show whether the relationship persists after accounting for these obligations.

Reference

Duchin, R., Ozbas, O. and Sensoy, B. A. (2010). Costly external finance, corporate investment, and the subprime mortgage credit crisis. *Journal of Financial Economics*, 97(3), 418–435. <https://doi.org/10.1016/j.jfineco.2009.12.008>

Appendix

1. I would like to investigate whether firms with more cash invest differently in machinery and equipment after Hurricane Helene. How can I turn this into a specific research question? Would it make sense to focus only on US manufacturing firms?
2. I have chosen Duchin, Ozbas and Sensoy (2010) as my reference. Their study focuses on the financial crisis, whereas mine concerns a hurricane. Can I apply this research to a shock of this kind or use it to inform my study?
3. What data do I need to examine capex and cash in the context of Helene? Should I use my own scraper for SEC filings in addition to Compustat?
4. You just suggested FEMA as a data source. What exactly is FEMA, what information does it provide, and how can this help me assess firms' exposure to Helene?
5. I would examine eight quarters before Helene and four full quarters afterwards. Should I exclude 2024Q3 because Helene occurred at the end of September?
6. I was planning to divide capex by current total assets. Could the investment ratio then rise simply because assets are destroyed by the storm? Would a fixed denominator from 2023 make sense? How would you implement this in practice?
7. What would a difference-in-differences model that tests my question about cash holdings look like? Please use Exposure, Post and Cash, along with firm and quarter fixed effects. Explain each term and use only the methods covered in our lecture slides.
8. Can I simplify the model without changing the research question?
9. I still do not understand the Post times Cash term. Why is the three-way interaction alone not enough? Please explain what change among unexposed firms we would otherwise miss.
10. Could firms with more cash also be better insured or better prepared? To what extent do firm fixed effects account for these differences?
11. As an extension, I would subtract short-term debt from cash holdings. How do I define this net liquidity measure using Compustat variables? Should I retain negative values and use the same sample and model for the comparison?
12. Please check my completed draft for inconsistencies between the research question, hypothesis, variables and regression. Where can I cut repetition so that the text fits within three pages? Suggest language improvements in English without adding new results or further references.